

Foundations and the Inclusion Exclusion Principle

The **sample space** or **universal set** has all the possible outcomes in the universe we’re working with, Ω . An **event** is a subset of the universal set. $A \subset \Omega$, and an event $x \in \Omega$ has occurred iff $x \in A$. The **complement** of an event is the event $A^C = \Omega \setminus A$.

Denote $\mathbb{A}$ as the set of all events.

Probability Measures

A Probability Measure is a map $\mathbb{P} : \mathbb{A} \rightarrow \mathbb{R}$ such that

- $\forall A \in \mathbb{A}, \mathbb{P}(A) \geq 0$
- $\mathbb{P}(\Omega) = 1$
- for countably many disjoint events A_i , (events such that $A_i \cap A_j = \emptyset$ when $i \neq j$), it holds that $\mathbb{P}\left(\bigcup_i A_i\right) = \sum \mathbb{P}(A_i)$

Discrete Uniform Distribution

If for all $w \in \Omega, \mathbb{P}(\{w\}) = \frac{1}{\# \Omega}$, then we call this a discrete uniform distribution, and $\mathbb{P}(A) = \frac{\#A}{\# \Omega}$

Combinatorics Basics

There are $n!$ ways to **order n objects**. n^k ways to **pull k objects with replacement**, and $\frac{n!}{(n-k)!}$ ways to pull k objects without replacement (but in order). There are $\binom{n}{k} = \frac{n!}{(k!)(n-k)!}$ if we ignore order (**n choose k**).

Inclusion-Exclusion Principle

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$
$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$$

Generally,

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}\left(\bigcap_{k=1}^k A_{i_j}\right)$$

Conditional Probability & Bayes Theorem

Conditional Prob.: $\mathbb{P}(B \mid A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)}$

Bayes: $\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A) \frac{\mathbb{P}(A)}{\mathbb{P}(B)}$

General Bayes: $P(A_j \mid B) = \frac{\mathbb{P}(B \mid A) \mathbb{P}(A_j)}{\sum_{k=1}^n \mathbb{P}(B \mid A_k) \mathbb{P}(A_k)}$

Two events are **independent** $\Leftrightarrow \mathbb{P}(A \mid B) = \mathbb{P}(A) \mathbb{P}(B)$

Random Variables

A **random variable (RV)** is a measurable variable on the sample space Ω . A random variable is a function $X : \Omega \rightarrow \mathbb{R}$, and is usually the results

of an experiment. If Ω is every possible theoretical outcome, X lets us work with Ω by taking real measurements.

A **cumulative distribution function (CDF)** is defined by $F_X(x) := \mathbb{P}(X \leq x)$ and makes sense for both discrete and continuous variables, but is usually used for continuous variables.

Discrete and Continuous RVs

If there exists some index set I that is at most countably infinite such that $\sum_{i \in I} \mathbb{P}(X = x_i) = 1$ we call X a discrete random variable. The **probability mass function** then is simply $p(x_i) = \mathbb{P}(X = x_i)$.

PDF: $\mathbb{P}(X \in B) = \int_B f_{X(x)} dx$

CDF: $P(X \leq B) = \int_{-\infty}^x f_X(t) dt$

Common Distributions

Distribution	PMF/PDF	$\mathbb{E}$	Var
Uniform on $\{1, ..., n\}$	$\mathbb{P}(X = k) = \frac{1}{n}$ for $k = 1, ..., n$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$
Bernoulli(p)	$\mathbb{P}(X = 1) = p$ $\mathbb{P}(X = 0) = 1 - p$	p	$p(1 - p)$
Binomial(n, p)	$\binom{n}{k} p^{k(1-p)^{n-k}}$ $k \in \{1, ..., n\}$	np	$np(1 - p)$
Geometric(p)	$\mathbb{P}(X = k) = p(1 - p)^{k-1}$ $k \in N$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	$e^{-\lambda} \frac{\lambda^k}{k!}$	λ	λ
Negative Binomial(r, p)	$\binom{k-1}{r-1} p^r (1 - p)^{k-r}$ $k \in \{r, r + 1, ... \}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Uniform([a, b])	$f_X = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Std Normal $N(0, 1)$	$f_X = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$	0	1
Gaussian $N(N, \sigma^2)$	$f_X = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Exponential(λ)	$f_X = \lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Uniform: All outcomes $1, ..., n$ are equally likely. **Bernoulli:** Success or failure, head or tail, yes or no. **Binomial:** How many successes there are out of a fixed number n of Bernoulli(p). **Geometric:** Counts number of trials till first success. **Possion:** Count of events / arrivals, e.g. calls in a call center, patients in a hospital. **Negative Binomial:** Trials until r-th success.

Uniform: Continuous uniform distribution on $[a, b]$, i.e. every x is equally likely ($\mathbb{1}_{[a,b]}(x)$ just checks if $x \in [a, b]$ and returns 1 if so else 0.) **Std.** **Normal:** Normal distribution with mean 0 and variance, e.g. standardized iid averages (**iid** = Independent and Identically Distributed.). **Gen.** **Normal:** General normal, usually assumed. **Exponential:** E.g. memoryless waiting times.

Two RVs are Independent

$\Leftrightarrow \mathbb{P}(X \in A \wedge Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B)$

Expected Value

$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx$

Expectations are linear: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$

Expectations add (even if not ind.): $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

Variance

$\text{Var } X = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

$\text{Var}(aX + b) = a^2 \text{Var}(X)$

Independence

X and Y are independent

$\Leftrightarrow \mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$

$\Leftrightarrow \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Linear and Monotone Transformations

If X, Y are continuous RVs, a, b are constants ($a \neq 0$), $g(x)$ is differentiable and strictly monotone function,

$Y = aX + B \Leftrightarrow f_Y = \frac{1}{|a|} f_X \left(\frac{y - b}{a} \right)$

$Y = g(X) \Leftrightarrow f_Y(y) = f_X(g^{-1}(y)) \cdot \left| (g^{-1})'(y) \right|$

Inverse function thm: $(g^{-1})'(y) = \frac{1}{g'(g^{-1}(y))}$

CDFs applied to their RVs are Uniform:

$Y = F_X(X) \Leftrightarrow Y \sim \text{Unif}([0, 1])$

Central Limit Theorem

Let $X_1, ..., X_n$ be iid random variables with $\mathbb{E}[X_1] = \mu \in \mathbb{R}$ and $\text{Var}(X_1) = \sigma^2 > 0$. We define $\overline{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then,

$\mathbb{P}\left(\frac{\overline{X}_n - \mathbb{E}[\overline{X}_n]}{\sqrt{\text{Var}(\overline{X}_n)}}\right) \xrightarrow{n \rightarrow \infty} \Phi(x)$

The CLT effectively states that for large N,

$\frac{\overline{X_n} - \mathbb{E}[\overline{X_n}]}{\sqrt{\text{Var}(\overline{X_n})}} \Big) \overset{\text{approx}}{\sim} N(0,1)$

Meaning we can estimate the sampling distribution from measurements (RVs), $X_1, ..., X_n$ using a standard normal curve!

Joint Probability Densities

If X, Y are independent, then $p_{X,Y}(x,y) = p_X(x)p_Y(y)$.

$$F_{X,Y} = \int_A \int_B f_{X,Y}(x,y)dydx$$

Covariance & Correlation

$$\text{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

X and Y are ind. $\implies \text{Cov}(X,Y) = 0$ (conv. not always true)

$$\text{Cov}(X,Y) = \text{Cov}(Y,X) \qquad \text{Cov}(X,X) = \text{Var}(X)$$

$$\text{Cov}(aX + bY + c, Z) = a \text{ Cov}(X, Z) + b \text{ Cov}(Y, Z) \forall a, b \in \mathbb{R}$$

$$\text{Correlation: Cor}(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X) \text{ Var}(Y)}}$$

$$-1 \leq \text{Cor}(X,Y) \leq 1$$

Cute Lemmas

$$\text{Prop 7.2.1 } \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{ Cov}(X,Y)$$

$$\text{Thm 7.22 } \text{Cov}(X,Y)^2 \leq \text{Var}(X) \text{ Var}(Y)$$

Useful Properties for Counterexamples

$$X = Y \text{ has } \text{Cov}(X,Y) = \text{Var}(X), \text{Cor}(X,Y) = 1,$$

$$\text{Var}(X + Y) = 4 \text{ Var}(X), \text{Var}(X - Y) = 0$$

$$Y = -X \text{ has } \text{Cor}(X,Y) = -1, \text{Var}(X + Y) = 0,$$

$$\text{Var}(X - Y) = 4\text{Var}(X)$$

Max of Ind. RVs

Let $X_i, i \in \{1, ..., n\}$ be iid RVs with cdf F_X . Then,

$$F_{\max_i X_i}(x) = F_X(x)^n$$

Min of Ind. RVs

Let $X_i, i \in \{1, ..., n\}$ be iid RVs with cdf F_X . Then,

$$F_{\min_i X_i}(x) = 1 - (1 - F_X(x))^n$$

Sum of RVs

$$p_{X+Y}(k) = \sum_y p_X(k-y)p_Y(y)$$

$$f_{X+Y}(z) = \int_{\mathbb{R}} f_X(z-y)f_Y(y)dy$$

Sum of Normal RVs

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

Sums of Distributions

$$X \sim \text{Pois}(\lambda_X), Y \sim \text{Pois}(\lambda_Y) \implies X + Y \sim \text{Pois}(\lambda_X + \lambda_Y)$$

$$X \sim \text{Binom}(n_X, p), Y \sim \text{Binom}(n_Y, p) \implies X + Y \sim \text{Binom}(n_X + n_Y, p)$$

$$X \sim \text{Exp}(\lambda), Y \sim \text{Exp}(\lambda) \implies X + Y \sim \text{Gamma}(2, \lambda)$$

$$X \sim \text{Gamma}(\alpha, \lambda), Y \sim \text{Gamma}(\beta, \lambda) \implies X + Y \sim \text{Gamma}(\alpha + \beta, \lambda)$$

$$X \sim \text{Exp}(\lambda_X), Y \sim \text{Exp}(\lambda_Y) \implies \min\{X, Y\} \sim \text{Exp}(\lambda_X + \lambda_Y)$$

Conditional Expectation

Let X be a discrete RV and A an event with $\mathbb{P}(A) > 0$, then

$$\mathbb{E}[X \mid A] = \sum_k x_k \mathbb{P}(X = x_k \mid A)$$

$$\text{Total Prob. for Exp. } \mathbb{E}[X] = \mathbb{E}[X|A_1]\mathbb{P}(A_1) + ... + \mathbb{E}[X|A_n]\mathbb{P}(A_n)$$

$$\text{Prop 9.3 } A = n \text{ is divisible by k, } B = n \text{ is divisible by l.}$$

$$A \text{ and } B \text{ are independent} \iff \text{gcd}(l, k) = 1$$

Chebyshev's Inequality

Let X be an RV with $E[X] = \mu$ and finite variance.

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Weak Law of Large Numbers

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of iid with finite variance. For any $a > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq a\right) = 0$$

Moment Generating Functions

Let X be an RV. Then, for $n \in \mathbb{N}$ we call $\mathbb{E}[X^n]$ the **nth-moment** of X . If $\mathbb{E}[X]$ is finite, then we further call $\mathbb{E}[(X - \mathbb{E}[X])^n]$ the **centered nth-moment**. This means the Expectation is the first moment and the Variance is the second centered moment.

The MGF of X, denoted by M_X is

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\left[\sum_{n=0}^{\infty} \frac{(tX)^n}{n!}\right]$$

$$\text{Cont. Form } M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x)dx$$

$$\text{Disc. Form } M_X(t) = \sum_x e^{tx} p_X(x)$$

To find the nth-moment, differentiate the MGF n-times, and evaluate at 0.

If $X_1, X_2, ..., X_n$ are independent RVs, then

$$M_{X_1+...+X_n}(t) = M_{X_1}(t) \cdot M_{X_2}(t) \cdot ... \cdot M_{X_n}(t)$$

Distribution	MGF
Uniform on $\{1, ..., n\}$	$\frac{e^t(e^{nt}-1)}{n(e^t-1)}$
Bernoulli(p)	$1 - p - pe^t$
Binomial(n, p)	$(1 - p + pe^t)^n$
Geometric(p)	$\frac{pe^t}{1-(1-p)e^t}$
Poisson(λ)	$\exp(\lambda(e^t - 1))$
Negative Binomial(r, p)	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$
Uniform([a, b])	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
Std Normal $N(0, 1)$	$\exp\left(\frac{t^2}{2}\right)$
Gaussian $N(N, \sigma^2)$	$\exp\left(\mu t + \sigma^2 \frac{t^2}{2}\right)$
Exponential(λ)	$\frac{\lambda}{\lambda-t}$

Brownian Motion

A **random walk** is a stochastic process $\{S_n : n \in \mathbb{N}_0\}$ defined by $S_0 = 0$ and $S_n = \sum_{i=1}^n X_i$, where $(X_i)_{i \geq 1}$ are RVs.

A **brownian motion** is a stochastic process, $\{B(t)\}_{t \in [0, \infty]}$, satisfying:

- $W(0) = 0$ (starts at 0)
- $W(t)$ has independent increments
- For $0 \leq s < t$, the increment $W(t) - W(s)$ is $N(0, t - s)$ distributed.
- $W(t)$ has continous paths (with probability 1).

Properties

- BM has almost surely continuous paths, but these paths are non-differentiable everywhere.
- BM is self-similar: for any constant $c > 0$, $(W(ct))_{t \geq 0}$ has the same distribution as $(\sqrt{c} \cdot W(t))_{t \geq 0}$.
- BM is a markov process, so the future depends only on the present.